

Final Presentation

# MedAire Inc.

Date: 05/02/2025



# Dynamic Rolling Forecast Model for MedAire Inc.



## Team 11

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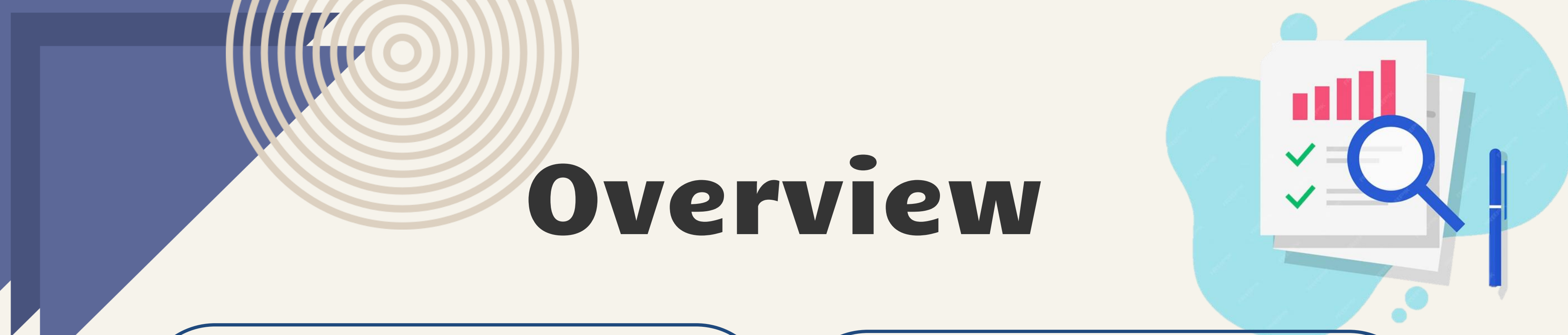
**Course:** MSBA 593 – Capstone  
Project



**Client:** Rob Miller, MedAire Inc.



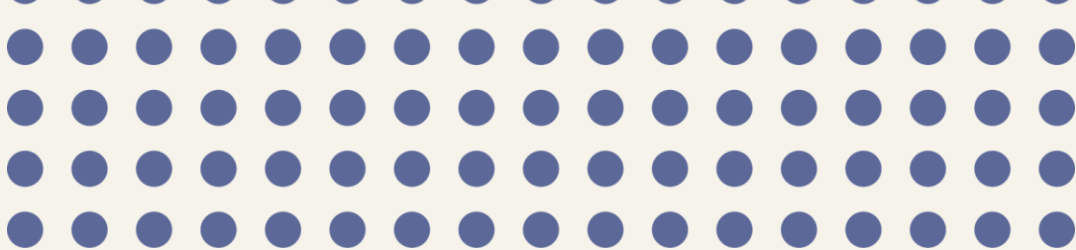
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# Overview

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# Problem Statement

- **Business Need:**
  - Forecast revenue across regions and products in a fast-changing market
  - Replace rigid static forecasts with a more dynamic, automated solution
- Traditional forecasting methods lack flexibility to react to real-time changes
- Revenue affected by:
  - Renewals (RNW), Do-Not-Renew (DNR)  
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  - Pricing changes
  - New business (clients/products)





# Project Objective

- Expand Phase 1 model (contractual) by integrating transactional revenue
- Clean and consolidate invoiced sales and P&L files
- Build automated forecast logic
- Design a Power BI dashboard for scenario-based planning

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# Tools

- **Excel:** Preliminary transformation and validation
- **Python:** Data wrangling, filtering, KPI alignment
- **Power BI:** Dynamic dashboards with filterable visuals
- **Data Sources:**
  - Invoiced Sales Data
  - Outlook EBIT (P&L)



# StakeHolder Meeting

- 18+ meetings with MedAire & Dr. Xiao
- Feedback guided changes:
  - Removed ML forecast in favor of **rule-based logic**
  - Ensured transparency & auditability in Excel and BI
- Shared files: FY 2425 A&M Inv Sales, Outlook EBIT, etc.

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# Project Progression

| Index    | Task                                         | Owner<br>(team member) | Start Date | End Date | Dependencies<br>(Task Indices) | Status*   |
|----------|----------------------------------------------|------------------------|------------|----------|--------------------------------|-----------|
| <b>1</b> | <b>Exploratory Data Analysis</b>             |                        |            |          |                                |           |
| 1.1      | Determine Data to Use for Model              | Dheeraj                | 01/21      | 02/04    |                                | Completed |
| 1.2      | Consolidating Data to One File               | Dominic                | 02/04      | 02/11    | 1.1                            | Completed |
| <b>2</b> | <b>Developing Model for Revenue Forecast</b> |                        |            |          |                                |           |
| 2.1      | Determine Formula for Revenue Calculation    | Dheeraj                | 02/11      | 02/18    |                                | Completed |
| 2.2      | Exploring Best ML Algorithm for Forecasting  | Ayush                  | 02/11      | 02/25    | 1.2                            | Completed |
| 2.3      | Company Cost Model                           | Riya                   | 02/11      | 02/25    | 1.2                            | Completed |
| 2.4      | Testing Validity of Model                    | Sravani                | 02/18      | 03/05    | 2.2 & 2.3                      | Completed |
| <b>3</b> | <b>PowerBI Dashboard Creation</b>            |                        |            |          |                                |           |
| 3.1      | Basic Layout & Design                        | Riya                   | 03/04      | 03/14    |                                | Completed |
| 3.2      | Formula Logic Implementation                 | Dheeraj                | 03/14      | 03/21    |                                | Completed |
| 3.3      | Consolidating Transactional Data             | Ayush                  | 03/21      | 03/28    |                                | Completed |
| 3.4      | Consolidating Contractual Data               | Dominic                | 03/21      | 03/28    |                                | Completed |
| <b>4</b> | <b>PowerBI Dashboard Implementation</b>      |                        |            |          |                                |           |
| 4.1      | Transactional Data Transformation How-To     | Ayush                  | 03/28      | 04/04    | 3.3                            | Completed |
| 4.2      | Contractual Data Transformation How-To       | Dominic                | 03/28      | 04/04    | 3.4                            | Completed |
| 4.3      | Fine-Tune Logic                              | Sravani                | 04/04      | 04/25    | 3.2                            | Completed |
| 4.4      | Final Design & Layout                        | Riya                   | 04/04      | 04/25    | 3.1                            | Completed |

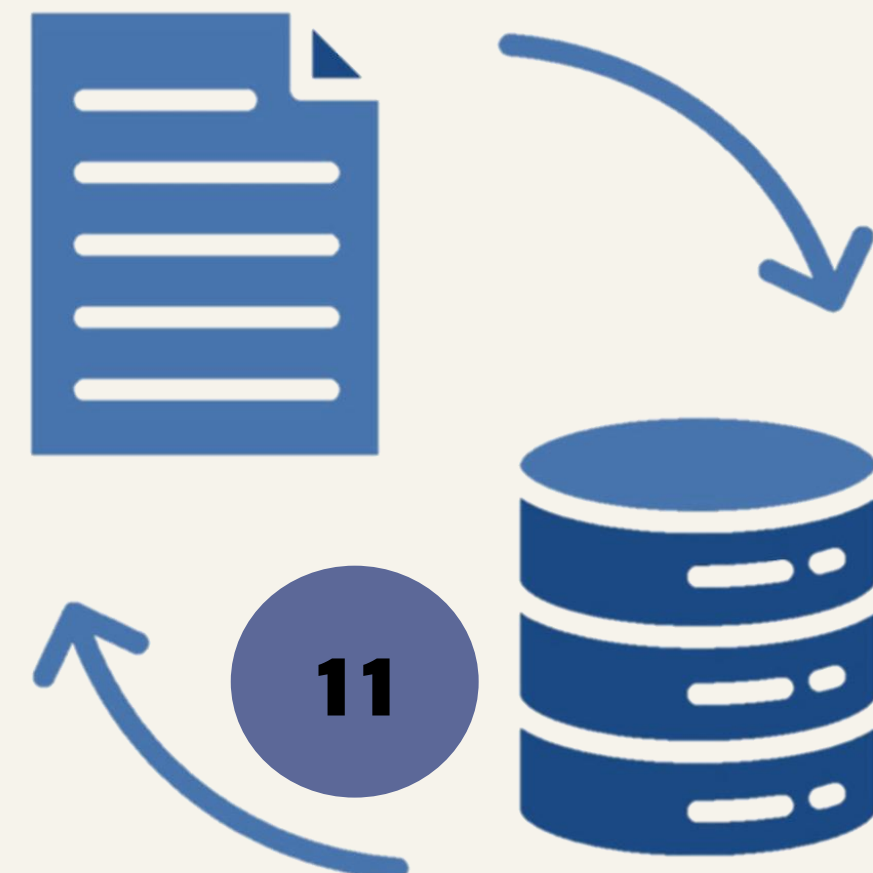
\*(Not Started, In Progress, Blocked, Completed)

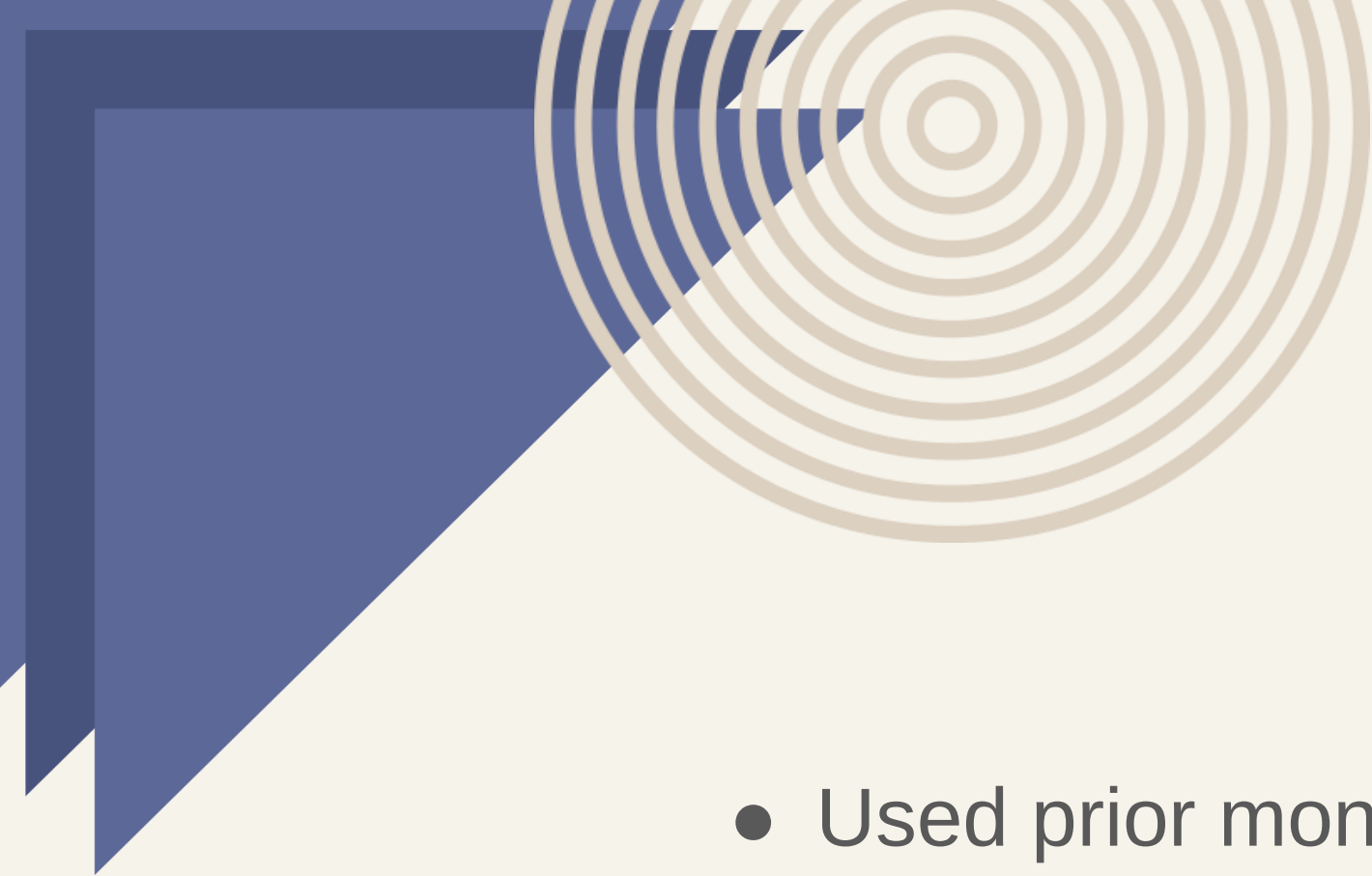


# Data Transformation

- Standardized date formats
- Tagged revenue types (contractual vs transactional)
- Aggregated monthly totals
- Implemented DNR, RNW, pricing % change logic
- Exported cleaned datasets to Power BI

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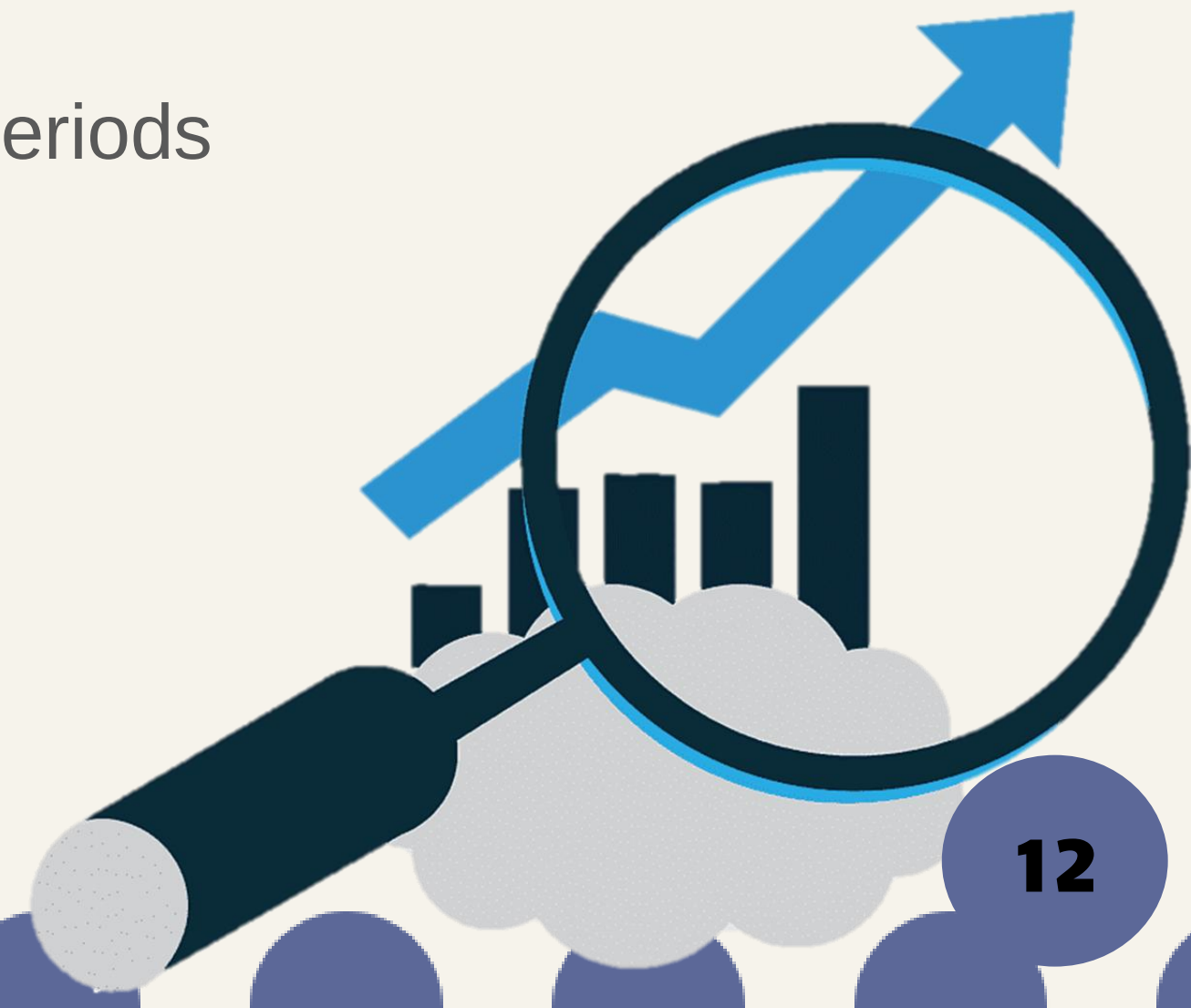


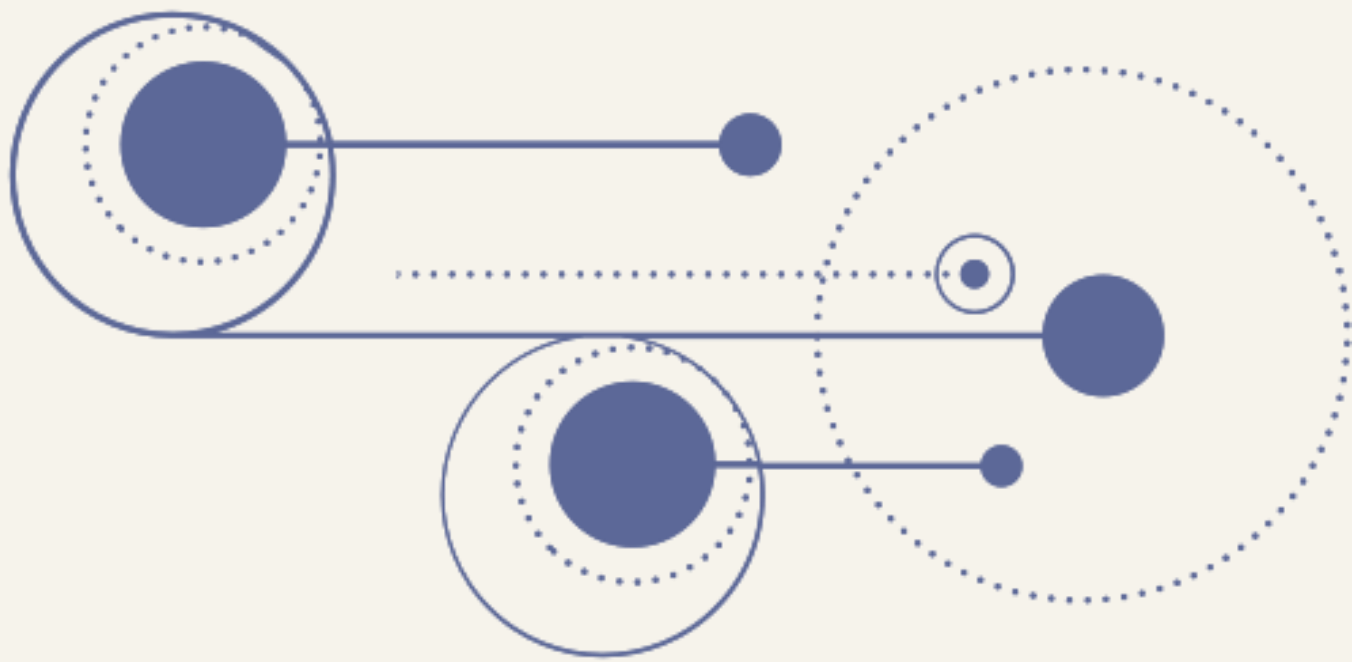


# Forecast Model

- Used prior month/year data as baseline
- Incorporated **levers**:
  - DNR%, Pricing %, New Clients, Refunds
- Recalculated renewals revenue over future periods
- Ensured month-by-month predictability

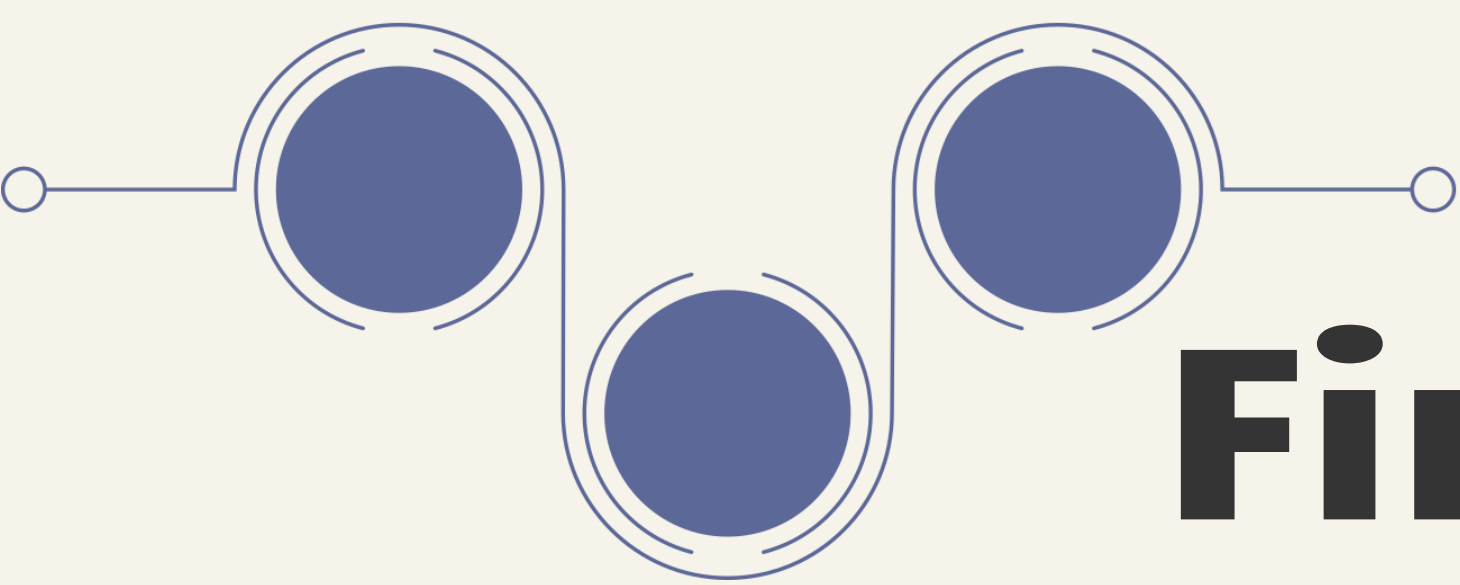
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# Key Insights/ Key Takeaways

- A forecast model was built to track monthly revenue from contracts and transactions.
- It showed patterns like seasonal changes and yearly growth.
- Transactional revenue changed more due to pricing and market shifts.
- Power BI dashboards<sup>11</sup> helped explore data by region, market, and product.
- Small changes (like more renewals or new products) had a big effect on revenue.
- The forecast matched past data well and showed how timing affected revenue.



# Final Deliverable

- Excel File - Data source for Power BI
- Python File - Data Cleaning code
- PowerBI Dashboard - EDA Dashboards ( Contractual & Transactional) and Overall Forecast Dashboard

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# Future Scope

- Automate ETL pipeline (Excel → Python → Power BI)
- Integrate forecasts with enterprise finance systems
- Apply predictive ML models when data stabilizes
- Extend dashboard to include cost, margin, and client risk scoring

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**MedAire®**  
An International SOS Company  
DIGITAL ASSESSMENT KIT

The image shows a black medical kit bag with the MedAire logo and text. The bag is placed on a white surface, possibly a table or bed. In the background, there are some medical supplies, including a white container and a blue bag. The overall scene is brightly lit, suggesting a clinical or emergency setting.

**THANK YOU!**

